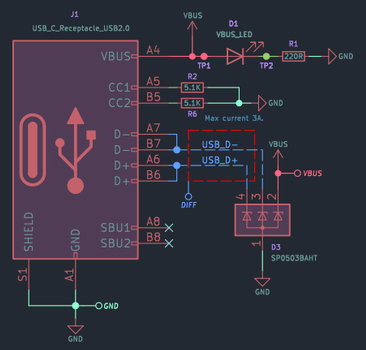
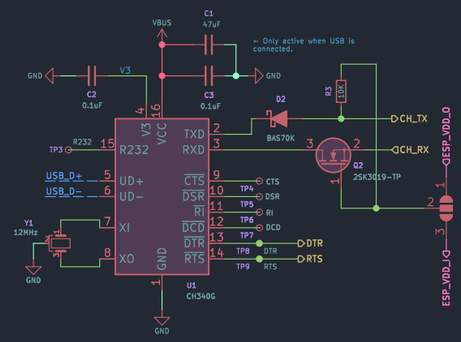
USB-C CONNECTOR

16-pin USB-C receptacle in USB 2.0 mode. Max current 3A.



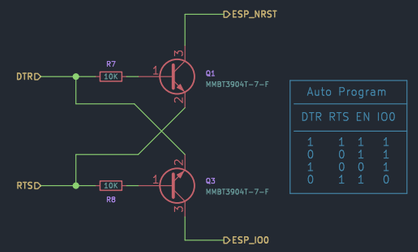
CH340G/C

USB 2.0 to serial converter.



AUTO-PROGRAM

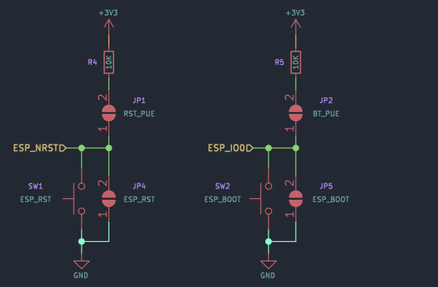
Auto reset and boot mode circuit. Uses DTR and RTS signals from the USB-Serial converter.



Auto Program			
DTR	RTS	EN	IO0
1	1	1	1
0	0	1	1
1	0	0	1
0	1	1	0

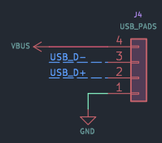
PUSH-BUTTONS

ESP reset and boot mode push-buttons for manual operation.
Also include solder points for external switches.



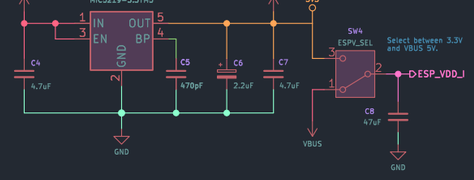
USB-C PADS

Allows soldering wires to the USB signals.



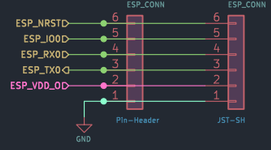
3.3V LDO

3.6-12V in, 3.3V out, 0.6V @ 500mA, PSRR 75dB (120Hz)



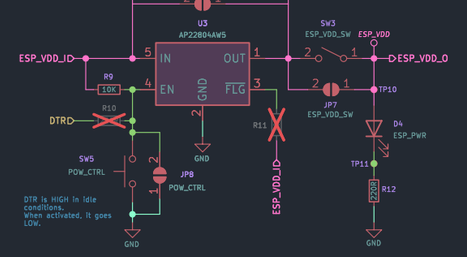
TARGET CONNECTORS

Generic connector for connecting to the target ESP.



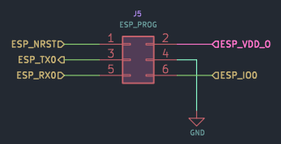
TARGET POWER CONTROL

Controls the power to the target through a load switch. You can also control the power through the DTR signal if it's connected.



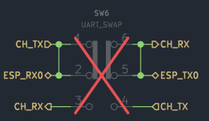
CONNECTOR – ESP-PROG

Allows to use the official ESP-Prog cables.



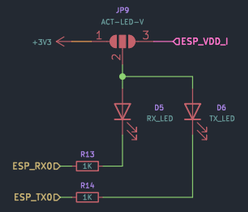
UART SWAP SWITCH

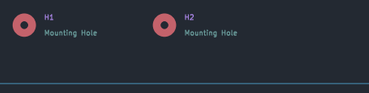
Swaps the TX-RX signals with a DPDT slide switch.



UART LEDs

Indicates activity on the UART lines.





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GitHub: https://github.com/CIRCUITSTATE/ESPNut-D1

Licence: MIT Open-Source Licence

ESP32 & ESP8266 Serial Auto-Programmer
Designed by: Vishnu Mohanan (@vishnumaiea, @vizmohanan)
CIRCUITSTATE Electronics LLP (www.circuitstate.com)

Sheet: /
File: ESPNut-D1.kicad_sch

Title: ESPNut-D1

Size: A3	Date: 2025-11-07	**Rev: 0.4**
KiCad E.D.A. 9.0.6		Id: 1/1